OPTIMA: Density-Aware Task Scheduling and Low-Latency IPC on Asymmetric Multiprocessors with Linux sched_ext

Anonymous Author(s) (Paper #184)

Abstract

Asymmetric chip-multiprocessors, combining high-throughput Performance (P) cores and power-efficient Efficiency (E) cores with disparate compute capacities and private cache topologies, present fundamental challenges for general-purpose operating system schedulers. While the Linux kernel incorporates capacity-aware scheduling (`sched-capacity`) and Energy-Aware Scheduling (EAS) to account for core compute differentials, standard fair-share classes (CFS, EEVDF) balance execution primarily by task virtual runtime lag and runnable load rather than execution density ($\rho = w/p$). Furthermore, existing schedulers lack a synchronous IPC bypass, inducing cross-CCX cache invalidation and high tail latencies when real-time and compute-bound tasks contend for heterogeneous execution resources.

In this paper, we present **OPTIMA**, an extensible operating system scheduler implemented in Linux `sched_ext` eBPF that provides principled algorithmic scheduling foundations for asymmetric multiprocessors. OPTIMA synthesizes three foundational mechanisms directly within the kernel hot-path: (1) a density-based adaptation of Smith's Rule (Weighted Shortest Processing Time, WSPT) that provably minimizes total weighted completion time $\sum w_j C_j$ on dedicated single-core queues and serves as an online approximation heuristic across asymmetric cores; (2) an Online Admissible Preemption Filter (OAPF) evaluating a dimensionally consistent one-step dispatch delay proxy $\Delta\text{DelayProxy}(c) = w_{\text{curr}}(c) \cdot (\tau_{\text{slice}} + C_{\text{switch}}) - w_{\text{wake}} \cdot R(c)$ in $\mathcal{O}(M_P)$ kernel time to suppress preemption thrashing and guarantee local delay reduction; and (3) a synchronous IPC zero-lag fast-path that routes waking communicating tasks to local dispatch queues, substantially reducing cross-CCX migrations (from 48.7% to 0.2%) and preserving L1/L2 cache locality. Using quadratic Lyapunov drift with an exact algebraic cross-term expansion, we formally prove that OPTIMA achieves conditional multi-queue stability under aggregate admissible capacity, while virtual deadline bounding ($v_j \leq t + 3\tau_{\max}$) protects batch tasks against starvation.

We evaluate OPTIMA across 16 diverse workloads (pro-audio DSP, high-frequency financial order matching, game physics server, API gateway fan-out, Redis caching, microbenchmarks, IPC ping-pong, and parallel Linux kernel compilation) against CFS/EEVDF, `scx_rusty`, and `scx_rlfifo`. OPTIMA exhibits a bimodal performance profile: delivering an **84.1% reduction in IPC ping-pong latency** (2.55 μs vs 16.06 μs on CFS) with a **6.3× throughput increase** (392,691 ops/s), a **93.6% increase in network loopback bandwidth** (111.47 Gbps), **zero audio underruns**, **zero game desyncs**, and a **94.6% reduction in worst-case real-time latency spikes**. Conversely, on parallel software compilation (`make kernel/`), OPTIMA takes 6.03s, matching other `sched_ext` baselines (5.67s–5.83s) while trailing native in-kernel CFS (3.29s) due to eBPF struct-ops trampoline overhead during fork/exec process storms, and shows moderate throughput regressions (14–30%) under high-concurrency request multiplexing (Redis, API Gateway). We present an in-depth architectural analysis characterizing these workload trade-offs and identifying the microarchitectural boundaries between specialized eBPF dispatch and in-kernel heuristics.

Keywords: Operating System Scheduling, eBPF, sched_ext, Asymmetric Multiprocessors, Online Preemption Filtering, WSPT, Lyapunov Stability, Real-Time Systems.

1 Introduction

The relentless demands of modern computing workloads—ranging from sub-millisecond audio digital signal processing and algorithmic trading to massively parallel software compilation and microservices—have driven microarchitectural design toward single-ISA asymmetric multiprocessors (AMP). Modern multi-core processors integrate heterogeneous core types: high-frequency, wide-issue Performance cores (P-cores) designed for single-thread burst performance, and high-density, power-efficient Efficiency cores (E-cores) designed to maximize parallel compute density. These cores typically feature disparate clock frequencies, asymmetrical instruction throughput per cycle (IPC), and non-uniform cache topologies organized into discrete core complex (CCX) clusters.

Despite these architectural advances, general-purpose operating system CPU scheduling remains rooted in symmetric multiprocessor (SMP) fair-share abstractions. Although Linux incorporates capacity-aware scheduling (`sched-capacity`) and Energy-Aware Scheduling (EAS) on supported platforms to track CPU capacity differentials, the default fair-share class (CFS [26] and EEVDF [36, 39]) balances execution primarily by task virtual runtime lag and runnable load rather than execution density ($\rho = w/p$). Consequently, existing kernel schedulers suffer from three distinct operational limitations under heterogeneous workloads:

1. **Capacity and Density Mismatch:** While capacity-

fitness checks migrate heavy tasks toward higher-capacity cores, they lack an execution-density metric to co-optimize priority weights against observed burst durations. High-priority interactive threads can experience capacity inversions or queueing delays behind batch jobs on Performance cores.

2. **Context-Switch Thrashing vs. Tail Latency:** When latency-critical tasks wake, CFS's virtual runtime preemption heuristics lack a dimensionally consistent preemption filter that accounts for context-switch and cache refill overheads, leading to either preemption storms or quantum wait spikes.

3. **Cross-Domain Cache Invalidation during Synchronous IPC:** In synchronous IPC and socket communication (e.g., pipe ping-pong, RPC fan-out), the default balancer searches for foreign idle cores across CCX boundaries, evicting hot L1/L2 caches and introducing costly inter-processor interrupt (IPI) latencies.

The recent introduction of `sched_ext` into upstream Linux 6.12 [11, 12] enables user-space and eBPF developers to replace the default kernel scheduling class with custom, dynamically loaded BPF struct-ops programs. While early sched-ext implementations, such as `scx_rusty` [12], `scx_lavd` [25], and the reference FIFO baseline `scx_rlfifo` [33], demonstrate the viability of eBPF scheduling, they rely primarily on heuristic balancing, user-space coordination daemons, or empirical policies that lack formal mathematical guarantees.

In this paper, we present **OPTIMA**, an extensible eBPF operating system scheduler that resolves these challenges through formal algorithmic synthesis. OPTIMA introduces mathematically grounded scheduling invariants directly into the kernel execution path:

- **Density-Based WSPT Sequencing (Smith's Rule):** We prove that ordering tasks by execution density $\rho_j = w_j/p_j$ is optimal for single-core dedicated queues ($1||\sum w_jC_j$), with ties permitted. While the general asymmetric problem with release dates ($R|r_j, \text{pmtn}|\sum w_jC_j$) is strongly NP-hard, OPTIMA leverages classical identical-machine results (Kawaguchi-Kyan reference bound) to formulate density-based virtual deadline scheduling as an online approximation heuristic, maximizing instantaneous backlog weight depletion via the Rearrangement Inequality.

- **Online Admissible Preemption Filtering (OAPF):** Rather than unconstrained heuristics, OPTIMA evaluates an online preemption filter over candidate Performance cores using the dimensionally consistent delay metric $\Delta\text{DelayProxy}(c) = w_{\text{curr}(c)} \cdot (\tau_{\text{slice}} + C_{\text{switch}}) - w_{\text{wake}} \cdot R(c)$. In $\mathcal{O}(M_P)$ bounded kernel time, OPTIMA provably preempts only when the one-step weighted delay is strictly reduced, suppressing preemption thrashing and enforcing bounded priority inversion ($\leq \tau_{\max}$).

- **Density-Guided Heterogeneous-Core Placement:** Core allocation between P- and E-cores is formulated via continuous fluid relaxation, preserving P-cores for high-density burst tasks and offloading long-running batch jobs ($p_j \geq 3$ ms) to Efficiency queues.

- **Synchronous IPC Zero-Lag Local Dispatch:** By intercepting synchronous wakeup flags (`SCX_WAKE_SYNC`) and dispatching communicating partners directly into local hardware runqueues (`SCX_DSQ_LOCAL`), OPTIMA achieves 2.55 μs IPC turnaround and 392k ops/s throughput.

- **Conditional Lyapunov Stability & Starvation Protection:** We prove that adaptive quantum scaling combined with work-conserving work-stealing guarantees negative Lyapunov drift under a per-queue service condition, while virtual deadline bounding ($v_j \leq t + 3\tau_{\max}$) protects batch tasks against starvation.

2 Related Work & Comparative Landscape

2.1 Extensible and User-Space Operating System Schedulers

Extensible operating system scheduling has a rich history. Early systems investigated microkernel-assisted and user-level scheduling to grant applications fine-grained control over execution priorities. Google ghOSt [13] demonstrated that delegating scheduling decisions to user-space daemons via shared-memory queues enables specialized microsecond-scale policies for warehouse workloads. However, ghOSt pays a communication penalty: cross-privilege context switches and thread yields incur 1–2 μs overheads.

Linux `sched_ext` [11, 12] overcomes this boundary by compiling verified BPF programs directly into kernel space. Early community schedulers explore distinct trade-offs: `scx_rusty` uses a hybrid model with a multi-threaded Rust user daemon coordinating BPF dispatch queues; `scx_lavd` [25] targets gaming and desktop interactivity through empirical critical-path deadline tracking; and `scx_rlfifo` [33] serves as a reference Round-Robin / FIFO baseline built on `scx_rustland_core`. OPTIMA differs fundamentally from these systems by enforcing formal combinatorial optimization algorithms directly in kernel bytecode without user-space daemon overhead.

2.2 General-Purpose Linux Kernel Scheduling

The Linux kernel has evolved through multiple scheduling paradigms. The Completely Fair Scheduler (CFS) [26] implemented Generalized Processor Sharing (GPS) [30] via red-black trees ordered by virtual runtime (v_{runtime}). While CFS guarantees asymptotic fairness on symmetric systems, its SMP load balancer relies on coarse heuristic averages that fail to account for asymmetric core speedups. Linux 6.6 introduced

EEVDF [39], replacing CFS's latency heuristics with Stoica and Abdel-Wahab's formal virtual deadline theory [36, 37]. EEVDF provides provable lag bounds on symmetric machines, but does not address asymmetric core partitioning or synchronous IPC cache locality.

For hard real-time workloads, `SCHED_DEADLINE` [21] implements Earliest Deadline First (EDF) coupled with the Constant Bandwidth Server (CBS) [1, 23]. While CBS guarantees temporal isolation, it requires static reservations (Q, P) and rejects unadmitted workloads. Kolivas's BFS [16] and MuQSS [17] highlighted that desktop latency spikes stem from heuristic load-balancing jitter, advocating for deterministic deadline ordering.

2.3 Asymmetric Multiprocessor (AMP) Scheduling

Scheduling on single-ISA heterogeneous processors was pioneered by Kumar et al. [19, 20]. Subsequent systems explored architectural bias scheduling [18] and contention-aware scheduling [2, 34, 38], demonstrating that memory-intensive tasks should be isolated to prevent cache degradation. Eyerman and Eeckhout [5, 6] analyzed CPI speedup stacks for asymmetric architectures.

On mobile platforms, ARM Energy-Aware Scheduling (EAS) [32] and DynamIQ evaluate energy differentials (ΔE) via an explicit kernel energy model. EAS prioritizes power reduction and reverts to standard CFS balancing under heavy load. Intel Thread Director [14, 29] provides hardware instruction classification via the Hardware Feedback Interface (HFI), but leaves policy enactment to the OS. In the domain of parallel runtimes, Chen et al. [4, 9] developed Selective Asymmetry-Aware Work-Stealing (SAWS) and Locality-Aware Work-Stealing (LAWS), showing that indiscriminate work-stealing across asymmetric cores degrades performance.

2.4 Theoretical Foundations of Scheduling

Smith's Rule [35] established that sequencing independent tasks in non-increasing order of density $\rho_i = w_i/p_i$ strictly minimizes total weighted completion time on an isolated single-core queue ($1||\sum w_i C_i$). However, in generalized multiprocessing, Lenstra et al. [22] and Garey and Johnson [7] proved that scheduling with dynamic release dates, preemption, and asymmetric parallel machines ($R|r_j, \text{pmtn}|\sum w_j C_j$) is strongly NP-hard (while Bruno et al. [3] proved NP-completeness for precedence-constrained tasks on identical processors). Classical approximation bounds, such as the Kawaguchi-Kyan bound ($\alpha \leq (1+\sqrt{2})/2 \approx 1.207$) [15], apply strictly to identical parallel machines ($P||\sum w_j C_j$) without release dates. Consequently, OPTIMA utilizes density-based list scheduling as a principled online approximation heuristic.

In online scheduling theory, Hall et al. [10], Megow and Schulz [24], and Phillips et al. [31] studied list scheduling guided by WSPT and linear programming relaxations. For preemption, rather than an intractable multi-stage branch-and-bound search tree, OPTIMA implements an Online Admissible Preemption Filter (OAPF) evaluated in $\mathcal{O}(M_P)$ kernel time. Finally, Neely's stochastic network optimization and Lyapunov drift analysis [8, 27, 28] provide the theoretical machinery to prove queue stability and bounded delay in stochastic queuing systems. Table 1 summarizes how OPTIMA advances beyond existing schedulers.

3 System Architecture & Implementation

OPTIMA is structured as a dual-plane operating system scheduler: a user-space telemetry and topology controller implemented in Rust, and a high-performance in-kernel scheduling engine implemented in eBPF via `sched_ext` struct-ops.

3.1 Heterogeneous Multi-Queue DSQ Hierarchy

Rather than maintaining a single global runqueue or uncoordinated per-CPU queues, OPTIMA partitions execution state across a hierarchical dispatch queue (DSQ) topology:

- `DSQ_PERF` (64-bit ID `0x100`): Dedicated exclusively to hard real-time and latency-critical tasks ($w_j \geq 500$ or pro-audio DSP). Executed with absolute priority by Performance cores.
- `DSQ_SHARED_P` (64-bit ID `0x101`): Shared domestic queue for general interactive and high-density tasks executing within the Performance core CCX domain.
- `DSQ_SHARED_E` (64-bit ID `0x102`): Shared domestic queue for balanced and throughput-oriented tasks executing within the Efficiency core CCX domain.
- `DSQ_EFF` (64-bit ID `0x103`): Long-running batch compute tasks ($p_j \geq 3\,\text{ms}$) demoted from P-cores.
- `SCX_DSQ_LOCAL`: Per-CPU hardware dispatch queue into which tasks are directly committed for immediate zero-latency dispatch without global lock contention.

3.2 Dynamic Topology Discovery

Upon loading, the user-space Rust controller inspects the sysfs microarchitectural hierarchy (`/sys/devices/system/cpu/cpu*/`). It extracts core frequency ceilings, physical package topologies, and hardware thread sibling maps. Cores are classified into Performance partition $\mathcal{P}$ and Efficiency partition $\mathcal{E}$. The controller initializes eBPF read-only global masks (`p_core_mask`, `e_core_mask`, `smt_sibling_map`), allowing the kernel BPF JIT compiler to perform dead-code elimination and optimize dispatch branches into constant bitwise operations.

Table 1: Comprehensive Comparison Matrix: OPTIMA vs. State-of-the-Art Operating System Schedulers

Scheduler	Asymmetry Awareness	Synchronous IPC Fast-Path	Preemption Provability	Tail-Latency Guarantees	Memory Footprint	Reconfig Overhead
OPTIMA *(This Work)*	Hardware-native multi-tier; dynamic sysfs topology detection; speedup $\mu = c_P/c_E$.	Direct `SCX_DSQ_LOCAL` on `SCX_WAKE_SYNC` (2.55 μs); SMT locality.	Online Admissible Preemption Filter (OAPF): $\Delta\text{DelayProxy} < 0$; bounded inversion ($\leq \tau_{\max}$).	Density-based WSPT + virtual deadlines; Lyapunov drift queue stabilization.	BPF `task_storage` (∼64 B); per-CPU array maps (∼16 KB total).	$\mathcal{O}(1)$ atomic BPF `.rodata` live tuning; lightweight user topology controller.
CFS / EEVDF *(Linux Default)*	Capacity-aware heuristics (`sched-capacity`); lacks execution density metrics.	Traverses full scheduling domain; cross-CCX IPI latency overhead.	Heuristic latency target (CFS); SMP lag bounds only (EEVDF).	Proportional lag bounded; susceptible to tail latency spikes under asymmetric contention and cross-domain load balancing.	Embedded in kernel `task_struct`; `sched_domain` tree.	Coarse sysctl tunables; requires kernel recompilation for logic changes.
Google ghOSt [13]	User-space policy dependent; lacks dynamic hardware speed models.	Cross-privilege boundary context switches (1–2 μs penalty).	Policy dependent; preemption delayed by user-agent wakeup.	Application-specific tail tuning; fragile under heavy preemption.	High; multi-MB shared memory queues and user thread pools.	Medium; requires updating user daemon or restarting agents.
scx_rusty [12]	Coarse domain clustering; does not compute dynamic core speedups.	BPF dispatch for local tasks; falls back to user daemon on imbalance.	Heuristic work-stealing based on queue depth ratios; no proof.	Moderate variance; tail spikes during user-space rebalancing.	Dual footprint: BPF maps plus Rust runtime memory heaps.	Low/Medium; CLI parameters modifiable via IPC channels.
scx_rlfifo [33]	Unaware; global shared queue without core differentiation.	Generic dispatch; lacks synchronous waker-affinity routing.	Round-robin / FIFO; periodic slice preemption without priority differentiation.	Susceptible to head-of-line blocking and severe tail spikes under mixed workloads.	Minimal BPF dispatch queue state (∼8 KB).	Static FIFO policy; requires restarting scheduler to modify parameters.
scx_lavd [25]	Topology-informed; tracks core frequencies for gaming interactivity.	Wakes latency-critical render threads on warm cores.	Heuristic critical-path tracking; lacks combinatorial bounds.	Low frame-time jitter; can penalize background batch throughput.	Low; in-kernel BPF maps for critical path tracking.	Low; live tuning of target frame rate deadlines.
ARM EAS [32]	Energy-model integrated; Capacity-Aware big.LITTLE scheduling.	Standard energy differential (ΔE) evaluation; no IPC bypass.	None; preemption driven by standard CFS heuristic rules.	Energy prioritized; defers to CFS load balancer under load.	Embedded in kernel source; static ACPI/device-tree tables.	Rigid; modifying models requires device-tree reconfiguration.

3.3 Synchronous IPC Zero-Lag Fast-Path

In synchronous client-server patterns (e.g., pipes, Redis GET/SET, microservice RPCs), the waking task and waker execute in lockstep. OPTIMA intercepts the `SCX_WAKE_SYNC` flag in `ops.select_cpu`. If the calling CPU (`this_cpu`) is allowed and has an empty local runqueue, OPTIMA immediately places the waking task into `SCX_DSQ_LOCAL` of `this_cpu`. If `this_cpu` is occupied, it attempts the SMT sibling (`this_sib`) or previous CPU (`prev_cpu`). This substantially reduces cross-CCX migrations (from 48.7% to 0.2%) and associated IPI interrupts, cutting ping-pong latency from 16.06 μs down to 2.55 μs.

3.4 Hierarchical Work-Conserving Work-Stealing

To prevent core starvation during parallel compute bursts (e.g., parallel software compilation), OPTIMA implements an asymmetric work-stealing protocol in `ops.dispatch`. Idle P-cores prioritize `DSQ_PERF` and `DSQ_SHARED_P`, but steal from `DSQ_SHARED_E` and `DSQ_EFF` when domestic queues are empty. Conversely, idle E-cores aggressively drain `DSQ_SHARED_P` without threshold barriers, preventing core underutilization and mitigating starvation under asymmetric compute bursts.

4 Mathematical Theory & Formal Guarantees

4.1 System Model and Problem Formulation

We consider an asymmetric multiprocessor system $\mathcal{M} = \mathcal{P} \cup \mathcal{E}$ with M_P Performance cores of capacity c_P and M_E Efficiency cores of capacity c_E, where $\mu = c_P/c_E > 1$. The aggregate compute capacity rate is:

$$C_{\text{total}} \triangleq M_P c_P + M_E c_E \tag{1}$$

Online tasks $\mathcal{J} = \{J_1, J_2, \dots\}$ arrive dynamically, specified by 5-tuples $J_j = (r_j, p_j, w_j, d_j, \mathcal{T}_j)$ with nominal processing requirement p_j (such that executing on core C_k with capacity c_k requires execution duration p_j/c_k), task density $\rho_j = w_j/p_j$, and priority weight $w_j \in [w_{\min}, w_{\max}]$ ($w_{\min} \geq 10$). The primary scheduling objective is to minimize total weighted completion time:

$$\min_{\mathcal{S}} \quad \Phi(\mathcal{S}) \triangleq \sum_{J_j \in \mathcal{J}} w_j C_j(\mathcal{S}) \tag{2}$$

subject to causality ($S_j \geq r_j$) and core exclusivity ($\sum_{J_j} \mathbf{1}_{\{J_j \text{ on } C_k \text{ at } t\}} \leq 1$).

Remark 1 (Latency-Sensitive Prioritization and Real-Time Scope)**.** *In an open, general-purpose operating system without hard admission control, arbitrary arrival bursts can exceed hardware processing capacity. Under unconstrained overload, guaranteeing hard real-time deadlines ($C_j \leq d_j$) for*

all tasks is mathematically impossible. OPTIMA provides latency-critical prioritization with zero-lag local dispatch (`SCX_DSQ_LOCAL`) and priority virtual deadlines ($v_j = t$) for latency-sensitive tasks ($w_j \geq w_{\mathrm{RT}}$), minimizing queueing delay under non-overloaded regimes, while providing graceful degradation under overload via density-weighted completion time heuristics.

4.2 Theorem 1: WSPT Sequencing, NP-Hardness, and Asymmetric Partitioning

Theorem 1 (Smith's Rule Optimality on Single-Core Dedicated Queues)**.** *Let $\mathcal{J}_k = \{J_1, \ldots, J_n\}$ be a set of n independent tasks available simultaneously at time t_0 ($r_j = t_0$) on an isolated processor core $C_k \in \mathcal{M}$. Any non-preemptive schedule ordering tasks in non-increasing density order:*

$$\rho_{(1)} \geq \rho_{(2)} \geq \cdots \geq \rho_{(n)} \tag{3}$$

minimizes the total weighted completion time $\sum_{j=1}^{n} w_j C_j$. (Arbitrary tie-breaking among tasks of equal density preserves identical objective value.)

Proof. Consider adjacent jobs J_a, J_b where J_a immediately precedes J_b with $\rho_a < \rho_b$, implying $w_a p_b < w_b p_a$. Interchanging them produces schedule $\mathcal{S}'$ without affecting any other tasks, yielding $\Delta\Phi = \frac{w_a p_b - w_b p_a}{c_k} < 0$. Adjacent pairwise exchanges strictly reduce total weighted completion time until jobs are sorted in monotonic non-increasing density order, establishing that Smith's ordering minimizes $\sum w_j C_j$ on an isolated single-core queue. □

Proposition 1 (Computational Complexity & Online Information Barrier)**.** *Minimizing total weighted completion time with dynamic release dates on parallel machines ($P|r_j|\sum w_j C_j$ or $R|r_j, pmtn|\sum w_j C_j$) is strongly NP-hard [7, 22]. Even on a single core, preemptive scheduling with arbitrary release dates ($1|r_j, pmtn|\sum w_j C_j$) is strongly NP-hard [22]. Furthermore, for online scheduling with unannounced arrivals, no online algorithm can achieve an optimal offline schedule regardless of computational power, as online decision-makers lack knowledge of future arrival sequences.*

Remark 2 (Context of the Kawaguchi-Kyan Bound)**.** *Kawaguchi and Kyan [15] proved that for identical parallel machines without release dates ($P||\sum w_j C_j$), WSPT list scheduling achieves an approximation factor of $\alpha \leq (1+\sqrt{2})/2 \approx 1.207$. This bound applies strictly to identical machines (P) at time zero. For asymmetric architectures (Q or R) and dynamic arrivals (r_j), OPTIMA uses density-based virtual deadline list scheduling as a principled online approximation heuristic.*

Theorem 2 (Continuous Fluid Relaxation for Density-Guided Placement)**.** *Under a continuous fluid relaxation where active jobs make continuous progress and priority weight is retired at a rate proportional to instantaneous compute bandwidth, the fluid weight retirement rate is $-\dot{W}_{\mathrm{fluid}}(t) = \sum_{k=1}^{M} \rho_{\sigma(k)} c_k$. By the Hardy-Littlewood-Pólya Rearrangement Inequality, this fluid retirement rate is maximized when tasks with higher density are matched monotonically to faster cores:*

$$\min_{k \in \mathcal{P}} \rho_{\sigma(k)} \geq \max_{k' \in \mathcal{E}} \rho_{\sigma(k')} \tag{4}$$

Definition 1 (Task Density, Reference Normalization ρ_{ref}, and Virtual Deadline Increment)**.** *For each task J_j with priority weight $w_j \in [w_{\min}, w_{\max}]$ ($w_{\min} \geq 10$) and estimated burst runtime $p_{\mathrm{est},j} \triangleq \max(\mathrm{avg_runtime}_j, \tau_{\min})$, its execution density is defined as $\rho_j \triangleq \frac{w_j}{p_{\mathrm{est},j}}$ with physical dimensions of $[\mathrm{weight}/\mathrm{time}]$ (or $[\mu\mathrm{s}^{-1}]$). To preserve dimensional consistency when translating density into temporal virtual deadline advances, we introduce an explicit reference density $\rho_{\mathrm{ref}} \triangleq \frac{w_{\mathrm{ref}}}{p_{\mathrm{ref}}} = \frac{100}{1000\,\mu\mathrm{s}} = 0.1\,\mu\mathrm{s}^{-1}$, representing a baseline interactive thread with default nice priority ($w_{\mathrm{ref}} = 100$) and a 1 ms execution burst ($p_{\mathrm{ref}} = 1000\,\mu\mathrm{s}$). The virtual deadline increment assigned during enqueue is:*

$$\begin{aligned} \Delta v_j &\triangleq \mathrm{clamp}\left(\tau_j \cdot \frac{\rho_{\mathrm{ref}}}{\rho_j},\ \tau_{\min},\ \tau_{\max}\right) \\ &= \mathrm{clamp}\left(\frac{\tau_j p_{\mathrm{est},j}}{10\max(w_j, 10)},\ \tau_{\min},\ \tau_{\max}\right) \end{aligned} \tag{5}$$

Because the normalized density ratio $\rho_{\mathrm{ref}}/\rho_j$ is strictly dimensionless, Δv_j has exact physical dimensions of time $[\mu\mathrm{s}]$, matching scheduling quantum τ_j.

Remark 3 (Decoupling Domain Placement from Intra-Queue Virtual-Deadline Ordering)**.** *We explicitly decouple coarse-grained domain placement from fine-grained intra-queue scheduling. First,* Domain Placement (Algorithm 2) *assigns tasks to target DSQs using heuristic thresholds: tasks with $w_j \geq w_{\mathrm{RT}}$ route to `DSQ_PERF`, long-running batch jobs ($p_{\mathrm{est},j} \geq 3\,\mathrm{ms}$ or `is_batch`) route to `DSQ_EFF`, and general interactive tasks route to domestic shared queues (`DSQ_SHARED_P` or `DSQ_SHARED_E`) preserving CPU stickiness; domain placement does not evaluate $\rho = w/p$ directly. Second,* Intra-Queue Scheduling (Algorithm 2) *evaluates task density ρ_j and reference density ρ_{ref} exclusively within the virtual deadline calculus ($\Delta v_j \propto \rho_{\mathrm{ref}}/\rho_j$) to establish the execution order ($\leq_v$) within shared dispatch queues, prioritizing higher-density tasks ahead of lower-density competitors.*

4.3 Theorem 3: Online Admissible Preemption Filtering (OAPF)

When a latency-critical task J_{wake} arrives and all P-cores are busy, OPTIMA evaluates an **Online Admissible Preemption Filter (OAPF)** greedy decision rule over candidate cores $c \in \mathcal{P}$ in $\mathcal{O}(M_P)$ kernel time, rather than an intractable multi-stage offline search tree.

Definition 2 (Weighted One-Step Dispatch Delay Proxy)**.** *Let $J_{\mathrm{curr}}(c)$ be the task executing on core $c \in \mathcal{P}$ with weight*

$w_{\text{curr}}(c)$ *and remaining quantum* $R(c)$*. Let waking task* J_{wake} *have weight* w_{wake} *and allocated time slice* $\tau_{\text{slice}} = \tau_{\min}$*. Let* C_{switch} *denote the context-switch and cache invalidation penalty in time units.*

If the scheduler waits, J_{wake} *is delayed by* $R(c)$*, incurring weighted dispatch delay* $w_{\text{wake}} \cdot R(c)$*. If the scheduler preempts,* $J_{\text{curr}}(c)$ *is suspended during* J_{wake}*'s slice (*τ_{slice}*) plus context-switch overhead* C_{switch}*, incurring weighted delay penalty* $w_{\text{curr}}(c) \cdot (\tau_{\text{slice}} + C_{\text{switch}})$*. The net differential in one-step weighted dispatch delay is:*

$$\Delta\text{DelayProxy}(c) \triangleq w_{\text{curr}}(c) \cdot (\tau_{\text{slice}} + C_{\text{switch}}) - w_{\text{wake}} \cdot R(c) \tag{6}$$

Both terms have unified physical dimensions of [weight $\times$ time] *(priority-weighted delay units).*

Theorem 3 (Local One-Step Dispatch Delay Reduction)**.** *Let* $\mathcal{P}_{\text{admissible}} \subseteq \mathcal{P}$ *be the set of candidate cores satisfying:*

1. *Priority superiority:* $w_{\text{wake}} > w_{\text{curr}}(c)$*;*
2. *Anti-thrashing remaining runtime:* $R(c) > \delta_{\min}(c)$*, where*

$$\delta_{\min}(c) \triangleq \frac{w_{\text{curr}}(c) \cdot (\tau_{\text{slice}} + C_{\text{switch}})}{w_{\text{wake}}} \tag{7}$$

If $\mathcal{P}_{\text{admissible}} \neq \emptyset$*, selecting core* $c^* = \arg\min_{c \in \mathcal{P}_{\text{admissible}}} \Delta\text{DelayProxy}(c)$ *and preempting* $J_{\text{curr}}(c^*)$ *guarantees that the one-step weighted dispatch delay is strictly reduced:*

$$\Delta\text{DelayProxy}(c^*) < 0 \tag{8}$$

Conversely, if $\mathcal{P}_{\text{admissible}} = \emptyset$ *or candidate cores satisfy* $R(c) \leq \delta_{\min}(c)$*, preemption is suppressed, bounding priority inversion to at most the running thread's quantum* $\tau(c) \leq \tau_{\max}$*.*

Proof. Under non-preemptive wait, J_{wake} incurs waiting delay $R(c^*)$, while $J_{\text{curr}}(c^*)$ incurs 0 delay, yielding weighted delay $\text{Delay}_{\text{wait}} = w_{\text{wake}} R(c^*)$. Under preemption, J_{wake} incurs 0 dispatch delay, while $J_{\text{curr}}(c^*)$ incurs delay $(\tau_{\text{slice}} + C_{\text{switch}})$, yielding weighted delay $\text{Delay}_{\text{preempt}} = w_{\text{curr}}(c^*)(\tau_{\text{slice}} + C_{\text{switch}})$. Computing $\Delta\text{DelayProxy}(c^*) = \text{Delay}_{\text{preempt}} - \text{Delay}_{\text{wait}}$ yields:

$$\Delta\text{DelayProxy}(c^*) = w_{\text{curr}}(c^*)(\tau_{\text{slice}} + C_{\text{switch}}) - w_{\text{wake}} R(c^*) \tag{9}$$

Since $c^* \in \mathcal{P}_{\text{admissible}}$, $R(c^*) > \delta_{\min}(c^*)$, which implies $w_{\text{wake}} R(c^*) > w_{\text{curr}}(c^*)(\tau_{\text{slice}} + C_{\text{switch}})$, ensuring $\Delta\text{DelayProxy}(c^*) < 0$.

When preemption is suppressed (e.g., $w_{\text{curr}}(c) \geq w_{\text{wake}}$ or $R(c) \leq \delta_{\min}(c)$), the waking task waits for quantum expiration. Because each executing slice is clamped to $\tau(c) \leq \tau_{\max}$, the maximum waiting delay is bounded by $\tau_{\max}$, guaranteeing bounded priority inversion while protecting cache locality and pipeline execution. □

Remark 4 (Tunable Anti-Thrashing Dampening Penalty $C_{\text{switch}} = 50\,\mu$s)**.** *While raw CPU register context switches require only* 1–2 μs *and individual cache line reloads require tens to hundreds of nanoseconds, repeatedly displacing and reloading a memory-intensive working set introduces cumulative pipeline stalls across thousands of cycles. In OPTIMA,* C_{switch} *serves strictly as a* tunable anti-thrashing dampening penalty *nominally set to* 50 μs *(with an effective operational sensitivity range of* 20–100 μs*). Rather than modeling a single hardware cache miss,* C_{switch} *enforces an algorithmic hysteresis barrier, ensuring that preemption is triggered only when the waking task's priority advantage is sufficiently pronounced to justify evicting the incumbent thread and disrupting pipeline throughput.*

4.4 OAPF-Adaptive: Peer Preemption Hysteresis

While Theorem 3 formally prevents lower-priority tasks from interrupting higher-priority incumbents, a naive interpretation where $w_{\text{curr}} \geq w_{\text{wake}}$ unconditionally prunes preemption introduces head-of-line (HoL) blocking in homogeneous-priority worker pools. In microservices, cache daemons (Redis), and benchmark harness workers (`schbench`), client and worker threads typically execute at identical default user priorities ($w_{\text{curr}} = w_{\text{wake}} = 100$). If an incumbent worker begins a lengthy compute burst up to $\tau_{\text{slice}} = 3\,\text{ms}$, newly arriving request threads with identical weights cannot preempt the running core under strict priority superiority ($w_{\text{wake}} > w_{\text{curr}}$), stalling in shared dispatch queues and inducing the multi-millisecond request P99 latency tails observed in Section 5.

To resolve this latency cliff without inducing context-switch thrashing, OPTIMA incorporates **OAPF-Adaptive**:

Definition 3 (Admissible Peer Preemption Criterion)**.** *Let* $J_{\text{curr}}(c)$ *and* J_{wake} *belong to equal priority classes (*$w_{\text{curr}}(c) = w_{\text{wake}}$*). Peer preemption is admissible if and only if the incumbent thread has attained forward execution progress of at least* $E(c) \geq \tau_{\text{soft}} = 1{,}000\,\mu$s *and possesses remaining quantum* $R(c) > \tau_{\min} = 500\,\mu$s*. The peer dispatch delay differential is defined as:*

$$\begin{aligned}\Delta\text{DelayPeer}(c) &\triangleq w_{\text{curr}}(c) \cdot C_{\text{switch}} - w_{\text{wake}} \cdot (R(c) - \tau_{\min}) \\ &= w \cdot \Big[C_{\text{switch}} - (R(c) - \tau_{\min})\Big]\end{aligned} \tag{10}$$

Core c^* *is preempted if* $\min_{c \in \mathcal{P}} \Delta\text{DelayPeer}(c) < 0$*.*

This hysteresis rule provides two vital guarantees:

1. **Anti-Thrashing Forward Progress:** Because $E(c) \geq 1{,}000\,\mu\text{s} \gg C_{\text{switch}} = 50\,\mu$s, the incumbent thread is guaranteed to have amortized its initial cache refill overhead before yielding, bounding the maximum switching frequency to at most 1 kHz per core.
2. **Head-of-Line Queue Delay Curtailment:** By preempting when remaining quantum exceeds $\tau_{\min}$, the maximum waiting delay experienced by peer request threads

is curtailed from $\tau_{\text{slice}} = 3\,\text{ms}$ down to $\leq 1\,\text{ms}$, directly mitigating request tail spikes in high-concurrency request pools.

4.5 Theorem 4 & Proposition 2: Conditional Queue Stability and Starvation Protection

We model the multi-queue system state vector $\mathbf{Q}(t) = (q_1(t), q_2(t), q_3(t), q_4(t))^T$ evolving via $q_k(t+1) = \max(0, q_k(t) - s_k(t)) + a_k(t)$ with bounded second moments $\mathbb{E}[a_k(t)^2 + s_k(t)^2 \mid \mathbf{Q}(t)] \leq M_{\text{sec}} < \infty$. We define the quadratic Lyapunov function:

$$L(\mathbf{Q}(t)) \triangleq \frac{1}{2} \sum_{k=1}^{4} q_k(t)^2 \tag{11}$$

Definition 4 (Aggregate Admissible Workload Region)**.** *The arrival workload is aggregate-admissible if aggregate workload demand is within total multiprocessor compute capacity:*

$$\sum_{k=1}^{4} \lambda_k \bar{p}_k \leq C_{\text{total}} - \epsilon_0 = (M_P c_P + M_E c_E) - \epsilon_0, \quad \epsilon_0 > 0 \tag{12}$$

where $\lambda_k = \mathbb{E}[a_k(t)]$ and $\bar{p}_k$ is mean task execution work.

Theorem 4 (Conditional Multi-Queue Stability via Foster-Lyapunov Drift)**.** *Let the multi-queue backlog $\mathbf{Q}(t)$ evolve under an aggregate-admissible workload (Definition 4). If the dispatch and hierarchical work-stealing policy guarantees a positive service slack across all active non-empty queues:*

$$\mathbb{E}\big[s_k(t) - a_k(t) \;\big|\; \mathbf{Q}(t)\big] \geq \delta > 0, \quad \forall k : q_k(t) > 0 \tag{13}$$

then the quadratic Lyapunov function satisfies the negative drift condition:

$$\Delta L(\mathbf{Q}(t)) \leq B - \delta \sum_{k=1}^{4} q_k(t) \tag{14}$$

for finite positive constant $B = \frac{1}{2}\sum_{k=1}^{4} \mathbb{E}[a_k(t)^2 + s_k(t)^2 \mid \mathbf{Q}(t)] < \infty$. Consequently, the multi-queue system is Foster-Lyapunov stable (positive recurrent), with bounded time-average queue depth:

$$\limsup_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} \sum_{k=1}^{4} \mathbb{E}[q_k(t)] \leq \frac{B}{\delta} < \infty \tag{15}$$

Proof. Expanding $q_k(t+1)^2$:

$$q_k(t+1)^2 = (\max(0, q_k(t) - s_k(t)))^2 + a_k(t)^2 + 2a_k(t)\max(0, q_k(t) - s_k(t)) \tag{16}$$

Using $(\max(0, x))^2 \leq x^2$, we bound the square by $(q_k - s_k)^2 = q_k^2 - 2q_k s_k + s_k^2$. For the cross-term, since $a_k \geq 0$ and $\max(0, q_k - s_k) \leq q_k$, we have $2a_k \max(0, q_k - s_k) \leq 2q_k a_k$. Combining both bounds:

$$q_k(t+1)^2 \leq q_k(t)^2 + a_k(t)^2 + s_k(t)^2 - 2q_k(t)\big(s_k(t) - a_k(t)\big) \tag{17}$$

Subtracting $q_k(t)^2$, dividing by 2, and taking conditional expectations:

$$\Delta L(\mathbf{Q}(t)) \leq B - \sum_{k=1}^{4} q_k(t)\, \mathbb{E}\big[s_k(t) - a_k(t) \;\big|\; \mathbf{Q}(t)\big] \tag{18}$$

Under the stated per-queue service condition (13), $\mathbb{E}[s_k(t) - a_k(t) \mid \mathbf{Q}(t)] \geq \delta > 0$ for non-empty queues. Substituting this into the drift bound yields $\Delta L(\mathbf{Q}(t)) \leq B - \delta \sum_k q_k(t)$. By the Foster-Lyapunov stability criterion, this establishes positive recurrence and strong aggregate queue stability. □

Remark 5 (Architectural Realization of the Service Slack Condition)**.** *In multi-queue systems with strict priorities, global capacity admissibility ($\sum \lambda_k \bar{p}_k \leq C_{\text{total}} - \epsilon_0$) is a necessary condition for aggregate stability, but does not independently ensure per-queue service slack (13) if high-priority queues can monopolize all cores. OPTIMA establishes this per-queue condition through structural partitioning: (1) dedicated Efficiency cores provide guaranteed domestic compute capacity to batch queues (`DSQ_EFF`), preventing high-priority P-core traffic from completely starving throughput work; and (2) dynamic quantum scaling $\tau(q_k) = \max(\frac{Q_{\text{base}}}{1+q_k}, \tau_{\min})$ dynamically compresses time slices during congestion, accelerating task dispatch and preemption frequency.*

Proposition 2 (Virtual Deadline Bounding and Starvation Protection)**.** *Under OPTIMA's virtual deadline calculus (Algorithm 2), for every task J_j enqueued into a batch or interactive queue (`DSQ_SHARED_P`, `DSQ_SHARED_E`, `DSQ_EFF`) at time t, its assigned virtual deadline satisfies:*

$$v_j(t) = \text{clamp}(v_{\text{time}}, t, t + 2\tau_j) + \Delta v_j \leq t + 3\tau_{\max} \tag{19}$$

since $\tau_j \leq \tau_{\max}$ and $\Delta v_j \leq \tau_{\max}$. Any subsequent batch task $J_{j'}$ arriving at timestamp $t' > t + 3\tau_{\max}$ receives $v_{j'}(t') \geq t' > v_j(t)$ and cannot be scheduled ahead of J_j. While a batch task arriving earlier ($t' \leq t + 3\tau_{\max}$) with higher density may receive an earlier deadline, $v_j(t) \leq t + 3\tau_{\max}$ strictly bounds relative queue advancement lag. Under the conditional service rate condition ($\mathbb{E}[s_k - a_k \mid \mathbf{Q}] \geq \delta > 0$) of Theorem 4, every admitted batch task is guaranteed dispatch within finite time. Furthermore, latency-critical tasks ($w \geq w_{\text{RT}}$) receive immediate priority ($v_j = t$) and always supersede batch tasks; when preemption is suppressed by OAPF, the immediate waiting delay imposed by the running task is bounded by $\tau_{\max}$, ensuring bounded priority inversion.

4.6 Algorithmic Specifications

Algorithms 1 and 2 detail the complete execution logic compiled into kernel eBPF bytecode.

Algorithm 1 OPTIMA CPU Selection & Online Admissible Preemption Filter (OAPF)

function OPTIMA-SELECT-CPU($p, prev_cpu, wake_flags$):
  $tctx \leftarrow$ lookup_task_ctx(p)
  if p is pinned $\vee$ p is kthread **then** dispatch $prev_cpu$ via `SCX_DSQ_LOCAL`; **return** $prev_cpu$
  $this_cpu \leftarrow$ smp_processor_id(), $idle_mask \leftarrow$ get_idle_smtmask()
  if $wake_flags$ & SCX_WAKE_SYNC **then**
    if $this_cpu \in p$.allowed $\wedge$ dsq_nr_queued($this_cpu$) $== 0$ **then goto** direct_dispatch with $this_cpu$
    $sib \leftarrow$ get_smt_sibling($this_cpu$)
    if $sib \in p$.allowed $\wedge$ dsq_nr_queued(sib) $== 0$ **then goto** direct_dispatch with sib
  if is_perf($prev_cpu$) $\wedge$ $prev_cpu \in idle_mask \wedge$ test_and_clear_idle($prev_cpu$) **then**
    goto direct_dispatch with $prev_cpu$
  if $tctx.core_type \neq$ CORE_EFF **then**
    $c \leftarrow$ pick_idle_domestic_core(p, is_perf $=$ TRUE, $idle_mask$)
    if $c \geq 0$ **then goto** direct_dispatch with c
  $is_ia \leftarrow (\neg tctx.is_batch \vee tctx.avg_runtime < 1\,\text{ms} \vee tctx.weight \geq w_{\text{RT}})$
  if $tctx.weight \geq w_{\text{RT}} \vee (is_ia \wedge$ dsq_nr_queued(`DSQ_SHARED_P`) $> 0)$ **then**
    $oapf \leftarrow$ OPTIMA-OAPF($p, tctx, \mathcal{P}$)
    if $oapf \geq 0$ **then** $tctx.dispatch_local \leftarrow$ TRUE; **return** $oapf$
  $tctx.dispatch_local \leftarrow$ FALSE; **return** $prev_cpu$
  direct_dispatch($target_cpu$): $\tau \leftarrow$ calc_slice($tctx$); scx_bpf_dsq_insert(p, SCX_DSQ_LOCAL, τ, 0); **return** $target_cpu$

function OPTIMA-OAPF($p, tctx, \mathcal{P}$):
  $c^* \leftarrow -1, \Delta\text{Proxy}^* \leftarrow 0, now \leftarrow$ scx_bpf_now()
  $w_{\text{wake}} \leftarrow \max(tctx.weight, 10), \tau_{\text{slice}} \leftarrow \tau_{\min}, C_{\text{switch}} \leftarrow 50\,\mu$s
  for each core $c \in \mathcal{P}$ **do**
    if $c \notin p$.allowed **then continue**
    $st \leftarrow$ lookup_cpu_run_state(c); **if** $st ==$ NULL $\vee$ $st.pid == 0$ **then return** c {Rule 1}
    $el \leftarrow now - st.start, R(c) \leftarrow (st.slice > el)\,?\,(st.slice - el) : 0,$ $w_{\text{curr}} \leftarrow \max(st.weight, 10)$
    if $w_{\text{curr}} == w_{\text{wake}}$ **then** {OAPF-Adaptive peer check}
      if $el \geq 1{,}000\,\mu$s $\wedge\, R(c) > 500\,\mu$s **then**
        lb $\leftarrow w_{\text{wake}} \cdot C_{\text{switch}} - w_{\text{wake}} \cdot (R(c) - 500)$
        if lb $< \Delta\text{Proxy}^*$ **then** $\Delta\text{Proxy}^* \leftarrow$ lb; $c^* \leftarrow c$
      continue
    if $w_{\text{curr}} > w_{\text{wake}}$ **then continue** {Rule 2: Inversion safety}
    $\delta_{\min} \leftarrow \frac{w_{\text{curr}} \cdot (\tau_{\text{slice}} + C_{\text{switch}})}{w_{\text{wake}}}$; **if** $R(c) \leq \delta_{\min}$ **then continue** {Rule 3}
    $\Delta\text{Proxy} \leftarrow w_{\text{curr}} \cdot (\tau_{\text{slice}} + C_{\text{switch}}) - w_{\text{wake}} \cdot R(c)$
    if $\Delta\text{Proxy} < \Delta\text{Proxy}^*$ **then** $\Delta\text{Proxy}^* \leftarrow \Delta\text{Proxy}; c^* \leftarrow c$
  if $c^* \geq 0 \wedge \Delta\text{Proxy}^* < 0$ **then** scx_bpf_kick_cpu(c^*, SCX_KICK_PREEMPT); **return** c^*
  return -1

Algorithm 2 OPTIMA-ENQUEUE & OPTIMA-DISPATCH

function OPTIMA-ENQUEUE(p, enq_flags):
  $tctx \leftarrow$ lookup_task_ctx(p)
  if p is pinned $\vee$ p is kthread $\vee$ $tctx.dispatch_local$ **then**
    scx_bpf_dsq_insert(p, SCX_DSQ_LOCAL, $\tau_{\min}, enq_flags$); **return**
  if $tctx.weight \geq w_{\text{RT}}$ **then** $target \leftarrow$ DSQ_PERF
  else if $tctx.avg_runtime \geq$ 3ms $\vee$ $tctx.is_batch$ **then** $target \leftarrow$ DSQ_EFF
  else $target \leftarrow$ is_perf(cpu) ? DSQ_SHARED_P : DSQ_SHARED_E
  $is_rt \leftarrow (tctx.weight \geq w_{\text{RT}} \vee (\neg tctx.is_batch \wedge tctx.avg_runtime < 1\text{ms}))$
  $now \leftarrow$ scx_bpf_now()
  if is_rt **then** $\tau \leftarrow \tau_{\min}$; $deadline \leftarrow now$
  else
    $\tau \leftarrow (tctx.avg_runtime \geq 3\text{ms})\,?\,\tau_{\max} : \max(\frac{Q_{\text{base}}}{1+q}, \tau_{\min})$
    $v \leftarrow \text{clamp}(p.\text{vtime}, now, now + 2\tau)$; $p_{\text{est}} \leftarrow \max(tctx.avg_runtime, \tau_{\min})$
    $\Delta v \leftarrow \text{clamp}\left(\frac{\tau \cdot p_{\text{est}}}{10 \cdot \max(tctx.weight, 10)}, \tau_{\min}, \tau_{\max}\right)$; $deadline \leftarrow v + \Delta v$
  $p.\text{vtime} \leftarrow deadline$
  scx_bpf_dsq_insert_vtime($p, target, \tau, deadline, enq_flags$)
  $idle \leftarrow$ scx_bpf_pick_idle_cpu(p.allowed, 0)
  if $idle \geq 0$ **then** scx_bpf_kick_cpu($idle$, SCX_KICK_IDLE); **return**
  $is_ia \leftarrow (\neg tctx.is_batch \vee tctx.avg_runtime < 1\text{ms} \vee tctx.weight \geq w_{\text{RT}})$
  if $is_rt \vee (is_ia \wedge$ dsq_nr_queued(`DSQ_SHARED_P`) $> 0)$ **then**
    $oapf \leftarrow$ OPTIMA-OAPF($p, tctx, \mathcal{P}$)
    if $oapf \geq 0$ **then** scx_bpf_kick_cpu($oapf$, SCX_KICK_PREEMPT)
function OPTIMA-DISPATCH(cpu):
  if is_perf(cpu) **then**
    for $dsq \in$ [PERF, SHARED_P, SHARED_E, EFF] **do**
      if scx_bpf_dsq_move_to_local(dsq, 0) **then return**
  else
    for $dsq \in$ [SHARED_E, EFF, SHARED_P, PERF] **do**
      if scx_bpf_dsq_move_to_local(dsq, 0) **then return**

5 Empirical Evaluation

5.1 Experimental Methodology & Hardware Microarchitecture

5.1.1 Testbed Microarchitecture: AMD Zen 5 / Zen 5c

All experiments are conducted on a bare-metal, asymmetric multi-core hardware platform powered by the **AMD Ryzen AI 9 HX 370 (Strix Point)** processor. The processor integrates 12 physical execution cores supporting 24 simultaneous hardware threads, architecturally partitioned into two asymmetric Core Complexes (CCXs):

- **Performance CCX (Zen 5):** 4 wide-issue Performance cores (Cores 0–3, threads 0–3 and 12–15) featuring a 6-wide decode execution pipeline, private 48 KB L1 data / 32 KB L1 instruction caches, 1 MB private L2 caches, and a shared 16 MB Level-3 cache slice, capable of boost clocks up to 5.16 GHz.
- **Efficiency CCX (Zen 5c):** 8 compact Efficiency cores (Cores 4–11, threads 4–11 and 16–23) utilizing area-optimized Zen 5c execution units, identical ISA support, private 1 MB L2 caches, and a dedicated 8 MB Level-3 cache slice, operating up to 3.29 GHz.

The compute capacity ratio yields an empirical single-thread compute speedup factor of $\mu = c_P/c_E \approx 1.57$ under frequency-normalized CPU throughput. The platform is equipped with 32 GB LPDDR5x-7500 dual-channel memory and runs Linux 6.12/7.0 with `CONFIG_SCHED_CLASS_EXT=y`. Throughout our evaluation, the baseline labeled "CFS / EEVDF" denotes the Linux 6.12 mainline default fair scheduler class (which implements Earliest Eligible Virtual Deadline First proportional-lag allocation with standard CFS load-balancing heuristics).

5.1.2 Execution Isolation & Hardware Energy Profiling

To eliminate external operating system noise, scheduling jitter, and background interference:

- **Isolated CPU Execution:** Workloads execute within an isolated 12-thread physical core partition (4 P-cores and 8 E-cores) with affinity pinning.
- **Dedicated IRQ Routing:** External device interrupts are affinity-pinned away from benchmark cores.
- **Hardware RAPL Energy Counters:** Package and core energy consumption is measured via Linux's unified `powercap/intel-rapl` MSR subsystem (which natively interfaces with AMD Zen 5 Family 0x1A RAPL

MSRs), tracking total Joules, average Watts, and Energy-Delay Product (EDP $=$ Joules $\times$ Time).

- **Statistical Rigor:** Every workload is executed through multiple independent repetitions (3 for performance benchmarks, 5 for RAPL energy). We report mean values, sample standard deviations ($\pm\sigma$), and relative percentage deltas.

5.2 Benchmark Workload Taxonomy

We evaluate 16 diverse workloads spanning four operational regimes: (1) *Real-Time & Multimedia* (Pro-Audio DSP turnaround, Game server physics ticks, and POSIX cyclic timer jitter); (2) *High-Concurrency Services* (High-Frequency Trading order matching, Production Redis in-memory cache, and Cloud API Gateway ingress fan-out); (3) *Microbenchmarks & IPC* (Perf sched pipe ping-pong, Perf messaging storm, Hackbench process/thread context-switching, and Iperf3 network loopback); and (4) *Compute & Memory Saturation* (Schbench worker wakeup/request scaling, Sysbench CPU multi-threaded compute, Sysbench memory transfer bandwidth, Hardware PMU L3 cache misses, Hardware instruction retirement IPC, and parallel Linux kernel compilation via `make kernel/`).

5.3 Performance Results

Table 2 reports the complete results across all 34 evaluated metrics for CFS/EEVDF, `scx_optima`, `scx_rusty`, and `scx_rlfifo`.

5.3.1 IPC Latency and Communication Throughput

In `perf bench sched pipe`, OPTIMA delivers an unprecedented **2.55 μs ping-pong latency**, representing an **84.1% latency reduction** compared to CFS (16.06 μs) and beating both `scx_rlfifo` (4.36 μs) and `scx_rusty` (6.19 μs). Throughput surges to **392,691 ops/s** (a **6.3$\times$ speedup** over CFS). This performance directly validates our synchronous IPC fast-path: by avoiding cross-CCX migration on `SCX_WAKE_SYNC`, cache lines remain hot in L1/L2 caches. In `iperf3` network loopback, OPTIMA achieves **111.47 Gbps**, nearly doubling CFS (57.58 Gbps) and outperforming `scx_rlfifo` (62.03 Gbps).

5.3.2 Real-Time Latency and Jitter Suppression

In POSIX real-time jitter evaluation (`cyclictest`), OPTIMA achieves the lowest average jitter (**21.27 μs**) and suppresses worst-case jitter spikes by **94.6%** (450 μs vs. 8,383 μs on CFS). In pro-audio DSP, OPTIMA achieves **31.71 μs** turnaround latency with **zero audio buffer underruns** (0.00 xruns vs. 3.67 on CFS). Similarly, in game server physics simulation, OPTIMA eliminates frame drops entirely (**0.000% desync rate**).

5.3.3 Software Compilation and Compute Throughput

By removing artificial work-stealing barriers and enabling idle E-cores to drain shared queues, OPTIMA achieves a kernel compile time of **6.03s**, closely matching peer extensible schedulers `scx_rusty` (5.67s) and `scx_rlfifo` (5.83s), though paying the expected BPF struct-ops trampoline tax compared to in-kernel CFS (3.29s). In `sysbench memory`, OPTIMA achieves highest-in-class bandwidth (**12,394.80 MiB/s**) and operation rates (**12,692,281 ops/s**), representing a **+12.8% gain** over CFS. Furthermore, thread fairness standard deviation σ drops by **98.0%** (46.48 vs. 2303.38 on CFS), demonstrating balanced cross-core workload distribution.

5.4 Ablation Studies

We conduct controlled ablation studies isolating (A) the synchronous IPC fast-path and (B) OAPF-Adaptive peer preemption.

Part A: Synchronous IPC Fast-Path: Disabling the synchronous IPC fast-path causes ping-pong latency to jump from 2.55 μs to 17.63 μs (matching CFS's 16.06 μs), while throughput collapses from 392,691 to 56,712 ops/s (-85.5%). Without immediate local enqueue, 48.7% of wakeups migrate across cache complex boundaries, triggering inter-processor interrupts (IPIs) and cache-line invalidation. With the fast-path enabled, communicating thread pairs execute back-to-back on the same physical core or SMT sibling, retaining hot cache lines and driving L1 data misses down from 8.42% to 1.15%.

Part B: OAPF-Adaptive Peer Preemption: As evaluated in Part B of Table 3, enabling OAPF-Adaptive dramatically curtails head-of-line request blocking under `schbench` worker contention. Request P99 tail latency drops by **72.6%** (from 170,752 μs down to 46,820 μs), request P50 drops by 32.2%, and aggregate service throughput surges by **+98.9%** (from 712.97 to 1,418.50 req/s), closely approaching CFS's 1,634 req/s. Concurrently, wakeup P99 jitter remains rock-solid (2,018.42 μs vs 2,007.33 μs), proving that OAPF-Adaptive successfully resolves homogeneous worker stalling without inducing context-switch thrashing.

6 Discussion and Limitations

Our comprehensive evaluation across 34 metrics reveals both significant microarchitectural strengths and principled trade-offs inherent to specialized eBPF scheduling on asymmetric multiprocessors. In this section, we analyze the underlying systems mechanisms governing these trade-offs and delineate the operational limitations of our approach.

6.1 Workload Trade-offs and Architectural Specialization

The empirical results demonstrate a clear bimodal performance profile:

Table 2: Comprehensive Benchmark Results Across 16 Workloads (34 Evaluated Metrics, Mean $\pm\sigma$)

Benchmark Workload	Metric Evaluated (Unit)	CFS / EEVDF	scx_optima	scx_rusty	scx_rlfifo	vs. CFS
Pro-Audio Turnaround Latency	Buffer P50 (μs) ↓	33.95 ± 4.28	$\mathbf{31.71 \pm 1.21}\star$	32.29 ± 1.69	33.86 ± 1.87	**-6.6%**
Pro-Audio Tail Latency	Buffer P99 (μs) ↓	101.50 ± 11.14	101.95 ± 2.02	61.91 ± 23.25	$\mathbf{58.42 \pm 21.62}\star$	+0.4%
Audio Buffer Underruns	Total Xruns (count) ↓	3.67 ± 0.58	$\mathbf{0.00 \pm 0.00}\star$	0.00 ± 0.00	0.00 ± 0.00	**0 drops**
Game Server Tick Latency	Physics P50 (μs) ↓	756.40 ± 146.34	777.14 ± 107.27	$\mathbf{690.73 \pm 35.76}\star$	718.47 ± 37.59	+2.7%
Game Server Frame Turnaround	Physics P99 (μs) ↓	$1{,}563.91 \pm 249.24$	$1{,}865.02 \pm 290.07$	$\mathbf{1{,}235.41 \pm 34.77}\star$	$1{,}387.65 \pm 150.65$	+19.3%
Game Server Frame Drops	Desync Rate (%) ↓	$0.019 \pm 0.03\%$	$\mathbf{0.000 \pm 0.00\%}\star$	$0.000 \pm 0.00\%$	$0.000 \pm 0.00\%$	**0.00% drops**
Real-Time POSIX Jitter	Avg Jitter (μs) ↓	24.17 ± 4.82	$\mathbf{21.27 \pm 4.68}\star$	23.90 ± 3.65	28.43 ± 5.55	**-12.0%**
Real-Time Jitter Max Spike	Worst Spike (μs) ↓	$8{,}383.00 \pm 8{,}269.70$	450.00 ± 339.69	468.33 ± 289.00	$\mathbf{280.67 \pm 23.12}\star$	**-94.6%**
HFT Order Matching Rate	Throughput (Mops/s) ↑	14.47 ± 7.28	15.76 ± 4.46	16.58 ± 3.83	$\mathbf{18.80 \pm 0.33}\star$	+8.9%
HFT Turnaround P50 Latency	Roundtrip P50 (μs) ↓	528.11 ± 587.94	$\mathbf{222.79 \pm 68.87}\star$	550.89 ± 515.18	244.76 ± 108.86	**-57.8%**
HFT Turnaround P99 Latency	Roundtrip P99 (μs) ↓	880.68 ± 962.33	576.34 ± 308.38	892.72 ± 487.04	$\mathbf{441.02 \pm 92.07}\star$	**-34.6%**
Production Redis Cache	GET Throughput (kRPS) ↑	$\mathbf{235.54 \pm 16.61}\star$	202.21 ± 4.16	186.44 ± 5.33	198.83 ± 14.28	-14.1%
Production Redis Cache	GET Latency P99 (μs) ↓	$\mathbf{249.67 \pm 32.33}\star$	319.00 ± 42.33	319.00 ± 42.33	399.00 ± 69.74	+27.8%
Cloud API Gateway Fan-Out	Ingress Rate (req/s) ↑	$\mathbf{1{,}275.61 \pm 157.17}\star$	887.95 ± 24.00	$1{,}066.03 \pm 128.06$	$1{,}001.99 \pm 62.50$	-30.4%
Cloud API Gateway Latency	Roundtrip P50 (ms) ↓	$\mathbf{33.83 \pm 4.42}\star$	49.58 ± 1.24	39.34 ± 3.52	41.75 ± 1.46	+46.6%
Cloud API Gateway Tail	Roundtrip P99 (ms) ↓	$\mathbf{76.97 \pm 17.69}\star$	100.78 ± 7.68	98.36 ± 18.55	108.00 ± 6.99	+30.9%
Hackbench Process Context-Switch	Process Time (s) ↓	0.35 ± 0.04	0.29 ± 0.01	0.37 ± 0.04	$\mathbf{0.21 \pm 0.03}\star$	**-16.9%**
Hackbench Thread Context-Switch	Thread Time (s) ↓	$\mathbf{0.60 \pm 0.02}\star$	0.67 ± 0.01	0.85 ± 0.04	0.85 ± 0.03	+12.3%
Perf Pipe IPC Latency	Ping-Pong Latency (μs) ↓	16.06 ± 0.65	$\mathbf{2.55 \pm 0.05}\star$	6.19 ± 4.12	4.36 ± 2.34	**-84.1%**
Perf Pipe IPC Throughput	Ping-Pong Rate (ops/s) ↑	$62{,}335 \pm 2{,}481$	$\mathbf{392{,}691 \pm 7{,}314}\star$	$205{,}677 \pm 100{,}235$	$269{,}291 \pm 110{,}723$	**+6.3** ×
Perf Messaging Storm	Wall Time (s) ↓	0.040 ± 0.003	0.045 ± 0.008	0.075 ± 0.031	$\mathbf{0.036 \pm 0.002}\star$	+11.7%
Schbench Wakeup P50 Latency	Wakeup P50 (μs) ↓	$\mathbf{10.00 \pm 9.54}\star$	$1{,}996.67 \pm 2.31$	$2{,}118.00 \pm 79.67$	$3{,}974.67 \pm 4.62$	+199.7 ×
Schbench Wakeup P99 Tail	Wakeup P99 (μs) ↓	$3{,}852.00 \pm 132.18$	$\mathbf{2{,}007.33 \pm 2.31}\star$	$15{,}472.00 \pm 400.96$	$6{,}077.33 \pm 92.38$	**-47.9%**
Schbench Request P50 Latency	Request P50 (μs) ↓	$\mathbf{18{,}698.67 \pm 1{,}420.07}\star$	$29{,}280.00 \pm 293.28$	$21{,}472.00 \pm 955.72$	$25{,}674.67 \pm 800.21$	+56.6%
Schbench Request P99 Tail	Request P99 (μs) ↓	$\mathbf{38{,}506.67 \pm 4{,}032.17}\star$	$170{,}752.00 \pm 5{,}701.39$	$284{,}842.67 \pm 14{,}201.27$	$140{,}373.33 \pm 9{,}473.15$	+343.4%
Schbench Service Rate	Throughput (req/s) ↑	$\mathbf{1{,}634.37 \pm 70.92}\star$	712.97 ± 7.33	714.00 ± 16.96	766.57 ± 23.47	-56.4%
Sysbench CPU Compute	Throughput (EPS) ↑	$\mathbf{11{,}712.23 \pm 72.58}\star$	$6{,}028.19 \pm 35.61$	$5{,}965.74 \pm 315.08$	$5{,}969.98 \pm 99.87$	-48.5%
Sysbench Thread Fairness	Std Dev σ (↓)	$2{,}303.38 \pm 286.35$	$\mathbf{46.48 \pm 15.65}\star$	223.38 ± 56.82	304.96 ± 65.14	**-98.0%**
Sysbench Memory Bandwidth	Throughput (MiB/s) ↑	$10{,}991.27 \pm 122.06$	$\mathbf{12{,}394.80 \pm 275.40}\star$	$12{,}376.37 \pm 598.71$	$11{,}891.51 \pm 101.28$	**+12.8%**
Sysbench Memory Operations	Operations Rate (Mops/s) ↑	11.26 ± 0.12	$\mathbf{12.69 \pm 0.28}\star$	12.67 ± 0.61	12.18 ± 0.10	**+12.8%**
L3 Cache Miss Rate	Hardware PMU (%) ↓	$6.40 \pm 2.41\%$	$2.66 \pm 0.09\%$	$1.55 \pm 0.04\%$	$\mathbf{1.36 \pm 0.06\%}\star$	**-58.5% Misses**
Hardware IPC Retirement	Instruction/Cyc (IPC) ↑	$\mathbf{1.18 \pm 0.00}\star$	0.88 ± 0.01	0.83 ± 0.00	0.83 ± 0.00	-25.2%
Network Loopback Throughput	iperf3 Rate (Gbps) ↑	57.58 ± 2.51	$\mathbf{111.47 \pm 5.19}\star$	38.23 ± 0.58	62.03 ± 1.85	**+93.6%**
Kernel Build (`make kernel/`)	Wall Time (s) ↓	$\mathbf{3.29 \pm 0.11}\star$	6.03 ± 0.08	5.67 ± 0.16	5.83 ± 0.06	+83.4%

Note: Values represent mean $\pm$ sample standard deviation (σ) across 3 independent executions. Star ($\star$) indicates best performance. Table 2 reports the baseline configuration with strict OAPF; Section 5.4 presents the OAPF-Adaptive peer-preemption ablation.

Table 3: Ablation Analysis: Synchronous IPC Fast-Path & OAPF-Adaptive Peer Preemption

Evaluated Metric	Ablated Baseline	OPTIMA Mode	Delta / Gain
Part A: Synchronous IPC Fast-Path Ablation			
Pipe Ping-Pong Latency (↓)	17.63 μs	**2.55 μs**	**-85.5% (6.9×)**
Pipe Ping-Pong Throughput (↑)	56,712 ops/s	**392,691 ops/s**	**+592.4%**
Network Loopback Rate (↑)	58.21 Gbps	**111.47 Gbps**	**+91.5%**
Cross-CCX Migrations (↓)	48.7%	**0.2%**	**-99.6%**
L1 Data Cache Miss Rate (↓)	8.42%	**1.15%**	**-86.3%**
Part B: OAPF-Adaptive Peer Preemption Ablation			
Schbench Wakeup P99 Tail (↓)	2,007.33 μs	2,018.42 μs	+0.5%
Schbench Request P50 (↓)	29,280 μs	**19,840 μs**	**-32.2%**
Schbench Request P99 Tail (↓)	170,752 μs	**46,820 μs**	**-72.6%**
Schbench Service Rate (↑)	712.97 req/s	**1,418.50 req/s**	**+98.9%**
Redis GET Throughput (↑)	202.21 kRPS	**228.45 kRPS**	**+13.0%**

- **Low-Latency IPC and Real-Time Robustness:** In workloads characterized by synchronous communication (`perf bench sched pipe`, `iperf3` socket loopback) and strict timing bounds (pro-audio DSP, game server ticks, POSIX jitter), OPTIMA achieves decisive improvements over CFS: an 84.1% reduction in pipe latency, a 94.6% reduction in jitter spikes, and zero audio underruns or frame drops. Relative to peer extensible schedulers, OPTIMA matches their zero-drop real-time robustness (0.00 xruns, 0.000% desync rate). In audio processing tail turnaround, OPTIMA achieves 101.95 μs P99 (closely tracking native CFS at 101.50 μs), while peer schedulers `scx_rusty` (61.91 μs) and `scx_rlfifo` (58.42 μs) demonstrate tighter tail bounds on this micro-benchmark. Concurrently, peer schedulers demonstrate advantages in specific isolated metrics: `scx_rusty` achieves faster game physics turnaround (1,235.41 μs P99 vs. OPTIMA's 1,865.02 μs and CFS's 1,563.91 μs), and `scx_rlfifo` registers the lowest peak jitter spike (280.67 μs vs. OPTIMA's 450.00 μs, though both represent a massive reduction over CFS's 8,383.00 μs spike). These variations reflect the trade-offs of distinct queue designs under pure real-time streaming.
- **High-Concurrency and Memory-Bound Trade-offs:** Conversely, in throughput-oriented database and proxy workloads (Production Redis, Cloud API Gateway), CFS outperforms OPTIMA by 14–30%. These applications exhibit thousands of short, non-blocking requests with highly volatile burst runtimes that challenge our EWMA burst estimation. Under rapid connection multiplexing, static or slow-converging EWMA values misclassify short batch tasks as compute-bound, intermittently routing them to Efficiency cores or incurring queue rebalancing delays.

6.2 Peer Preemption, OAPF-Adaptive, and Request Latency Trade-offs

A notable finding in Table 2 is the behavior under `schbench`: OPTIMA's wakeup P50 latency is $1{,}996.67 \pm 2.31$ μs compared to CFS's 10.00 ± 9.54 μs. Importantly, this $\sim$2 ms P50

latency floor is shared across all evaluated `sched_ext` schedulers (`scx_rusty` at $2{,}118.00 \pm 79.67\,\mu$s and `scx_rlfifo` at $3{,}974.67 \pm 4.62\,\mu$s). This phenomenon is an intrinsic consequence of dispatch queue scheduling in `sched_ext`: newly woken threads await executing threads' time slices to elapse before being dispatched from DSQs, whereas native CFS preempts executing threads nearly instantaneously ($10\,\mu$s) based on virtual runtime lag (v_{runtime}).

However, under heavy request concurrency, aggressive peer preemption under CFS causes cascading context switches, expanding CFS's wakeup P99 tail to $3{,}852.00 \pm 132.18\,\mu$s and causing `scx_rusty`'s tail to explode to $15{,}472.00 \pm 400.96\,\mu$s. In contrast, standard OPTIMA holds its wakeup P99 tail down to $\mathbf{2{,}007.33 \pm 2.31}\,\mu$s—a remarkably tight distribution differing from P50 by only $10.66\,\mu$s. This stability is governed directly by **Rule 2 of the Online Admissible Preemption Filter (OAPF)**:

$$w_{\text{curr}}(c) \geq w_{\text{wake}} \implies \text{Prune candidate } c$$

Because all `schbench` worker threads share identical nice levels ($w = 100$), standard Rule 2 intentionally suppresses peer preemption among equal-priority threads. By guaranteeing that once a thread begins its execution quantum ($\tau_{\min} = 2\,$ms) it runs without interruption, OPTIMA eliminates thrashing and achieves rock-solid wakeup tail latency predictability.

Nonetheless, this anti-thrashing design imposes a substantial trade-off on end-to-end request-response metrics under worker pool contention. When worker threads cannot preempt peers, queued request-handling tasks must wait for running threads to yield or exhaust full quanta. Consequently, OPTIMA's request P50 latency increases to $29{,}280.00 \pm 293.28\,\mu$s (+56.6% vs. CFS's $18{,}698.67 \pm 1{,}420.07\,\mu$s), request P99 tail latency rises to $170{,}752.00 \pm 5{,}701.39\,\mu$s (+343.4% vs. CFS's $38{,}506.67 \pm 4{,}032.17\,\mu$s, though outperforming `scx_rusty`'s $284{,}842.67 \pm 14{,}201.27\,\mu$s), and aggregate service throughput declines to 712.97 ± 7.33 req/s (-56.4% vs. CFS's $1{,}634.37 \pm 70.92$ req/s).

OAPF-Adaptive Remediation: To eliminate this head-of-line blocking while retaining thrashing resistance, Section 4.4 introduced OAPF-Adaptive. By allowing soft peer preemption after an incumbent thread has completed $\tau_{\text{soft}} = 1\,$ms of progress and possesses remaining slice $R(c) > 500\,\mu$s, peer request threads are no longer starved behind 3 ms quantum monopolizations. This bounds head-of-line delay to $\leq 1\,$ms, bridging the gap between predictable wakeup jitter and high-concurrency request throughput in homogeneous server pools.

6.3 Sysbench Compute Throughput and Thread Fairness

In `sysbench cpu`, all three `sched_ext` schedulers cluster near ∼6,000 EPS (OPTIMA: $6{,}028.19 \pm 35.61$ EPS, `scx_rusty`: $5{,}965.74 \pm 315.08$ EPS, `scx_rlfifo`: $5{,}969.98 \pm 99.87$ EPS) within the 12-thread isolated partition, whereas CFS achieves $11{,}712.23 \pm 72.58$ EPS. This occurs because the pure compute loop of sysbench creates continuous core contention where CFS allows threads to unrestrictedly saturate all core pipelines without queue structure overheads. However, the critical differentiator lies in thread fairness: under CFS, individual thread event completions exhibit extreme variance ($\sigma = 2{,}303.38$), with some threads starved on slower cores while others surge ahead. Under OPTIMA, per-thread standard deviation drops by **98.0%** down to $\sigma = 46.48 \pm 15.65$. This demonstrates that OPTIMA's density-guided domain placement ensures uniform compute progress across asymmetric cores at the expense of uncoordinated thread racing.

6.4 Extensible Scheduler Overheads in Dynamic Process Lifecycles

In the parallel Linux kernel compilation benchmark (`make kernel/`), the wall-clock build time is $6.03 \pm 0.08\,$s for OPTIMA, matching `scx_rusty` ($5.67 \pm 0.16\,$s) and `scx_rlfifo` ($5.83 \pm 0.06\,$s), but lagging native CFS ($3.29 \pm 0.11\,$s). In earlier unoptimized iterations of OPTIMA, build time was 11.63 s due to an artificial queue barrier that prevented idle Efficiency cores from stealing tasks enqueued for Performance cores. Removing this barrier and enabling unconstrained asymmetric work-stealing reduced compilation time by **48.2%** (11.63 s → 6.03 s).

However, all `sched_ext` schedulers remain ∼83% slower than CFS on this workload. The root cause is microarchitectural trampoline overhead: kernel compilation executes tens of thousands of short-lived compiler processes (`gcc`, `as`, `ld`). For each process creation and termination, `sched_ext` invokes BPF struct-ops hooks (`ops.init_task`, `ops.exit_task`, `ops.select_cpu`, `ops.enqueue`). While each invocation costs only a few hundred nanoseconds, the cumulative overhead across hundreds of thousands of process lifecycles imposes a measurable penalty compared to in-kernel CFS C code that optimizes process creation paths natively.

6.5 Microarchitectural Asymmetry and Hardware IPC Retirement

In Table 2, OPTIMA registers a hardware IPC retirement of 0.88 ± 0.01 (a -25.2% delta vs. CFS at 1.18 ± 0.00). Importantly, peer extensible schedulers also exhibit lower retirement rates (both `scx_rusty` and `scx_rlfifo` register 0.83 ± 0.00 IPC), demonstrating that this lower instruction retirement is not an isolated side effect of OPTIMA's density steering, but rather reflects general microarchitectural execution characteristics and memory-bound stall patterns across the benchmark suite under `sched_ext`. In OPTIMA, steering batch compute to narrower Efficiency pipelines does contribute to lower per-cycle retirement for those specific threads, but OPTIMA actually achieves slightly higher retirement (0.88) than peer schedulers (0.83) due to improved cache locality

Table 4: Hardware RAPL Energy Profiling & Energy-Delay Product (AMD Ryzen AI 9 HX 370, 5 Repetitions)

Energy Metric (Unit)	CFS / EEVDF	scx_optima	scx_rusty	scx_rlfifo	vs. CFS
Sysbench CPU Compute (12 threads, 5.0s window)					
Package Energy (J) ↓	180.89 ± 1.49	179.68 ± 1.52	$\mathbf{176.27 \pm 0.64}\star$	176.68 ± 0.50	-0.7%
Core Energy (J) ↓	33.69 ± 0.59	34.13 ± 0.31	33.01 ± 0.64	$\mathbf{33.00 \pm 0.50}\star$	+1.3%
Avg Package Power (W)	36.14 ± 0.30	35.90 ± 0.30	$\mathbf{35.22 \pm 0.13}\star$	35.30 ± 0.10	-0.7%
EDP (J·s) ↓	905.39 ± 7.43	899.28 ± 7.65	$\mathbf{882.22 \pm 3.10}\star$	884.36 ± 2.63	-0.7%
Sysbench Memory Bandwidth (12 threads, 5.0s window)					
Package Energy (J) ↓	137.69 ± 2.91	138.60 ± 2.02	$\mathbf{131.24 \pm 0.56}\star$	133.87 ± 0.78	+0.7%
Core Energy (J) ↓	22.29 ± 0.29	23.32 ± 0.58	$\mathbf{21.87 \pm 0.26}\star$	22.25 ± 0.12	+4.6%
Avg Package Power (W)	27.51 ± 0.58	27.69 ± 0.41	$\mathbf{26.22 \pm 0.11}\star$	26.75 ± 0.16	+0.7%
EDP (J·s) ↓	689.12 ± 14.52	693.74 ± 10.10	$\mathbf{656.85 \pm 2.85}\star$	670.02 ± 3.91	+0.7%
IPC Pipe Ping-Pong (100k operations)					
Package Energy (J) ↓	4.01 ± 0.62	$\mathbf{3.10 \pm 0.17}\star$	3.44 ± 0.20	3.54 ± 0.39	**-22.7%**
Core Energy (J) ↓	0.96 ± 0.18	0.71 ± 0.51	0.91 ± 0.68	$\mathbf{0.58 \pm 0.52}\star$	**-26.0%**
Avg Package Power (W)	28.71 ± 2.49	22.75 ± 1.32	21.10 ± 4.28	$\mathbf{19.00 \pm 3.40}\star$	**-20.8%**
EDP (J·s) ↓	0.58 ± 0.19	$\mathbf{0.42 \pm 0.02}\star$	0.57 ± 0.08	0.67 ± 0.15	**-27.6%**
Hackbench Context-Switch (isolated process group)					
Package Energy (J) ↓	6.73 ± 0.78	4.12 ± 0.18	9.62 ± 1.58	$\mathbf{4.00 \pm 0.16}\star$	**-38.8%**
Core Energy (J) ↓	1.14 ± 0.11	0.85 ± 0.03	1.70 ± 0.28	$\mathbf{0.69 \pm 0.04}\star$	**-25.4%**
Avg Package Power (W)	45.34 ± 0.87	48.05 ± 1.16	$\mathbf{38.06 \pm 1.95}\star$	46.76 ± 0.89	+6.0%
EDP (J·s) ↓	1.01 ± 0.23	$\mathbf{0.35 \pm 0.04}\star$	2.53 ± 1.03	$\mathbf{0.34 \pm 0.03}\star$	**-65.3%**

Note: Mean $\pm\sigma$ across 5 runs via `powercap/intel-rapl` MSRs on AMD Zen 5/5c. Star ($\star$) indicates lowest energy/EDP.

(reducing L3 cache misses by 58.5% down to $2.66 \pm 0.09\%$ vs CFS's $6.40 \pm 2.41\%$).

Notably, in Table 2, `scx_rlfifo` registers an even lower L3 miss rate ($1.36 \pm 0.06\%$ vs. $2.66 \pm 0.09\%$ on OPTIMA and $6.40 \pm 2.41\%$ on CFS). This occurs because `scx_rlfifo`'s flat, single-queue FIFO design preserves strictly sequential task execution without domain rebalancing or cross-queue work-stealing passes, whereas OPTIMA incurs a minor cache footprint from hierarchical DSQ classification and work-stealing checks across the asymmetric topology.

6.6 Hardware RAPL Energy Efficiency & Energy-Delay Product (EDP)

Asymmetric multi-core processors were fundamentally engineered to optimize energy efficiency by offloading long-running non-interactive workloads to dense, low-power execution units.

Table 4 presents hardware RAPL energy measurements collected across 5 independent runs on the host AMD Zen 5/5c processor via Linux's unified `powercap/intel-rapl` driver (which natively interfaces with AMD Zen 5 Family 0x1A RAPL MSRs):

- **Synchronous IPC Energy Dividends:** In fixed-work `perf bench sched pipe` (100k operations), OPTIMA reduces total package energy from 4.01 ± 0.62 J on CFS down to $\mathbf{3.10 \pm 0.17}$ J (**-22.7%**), while core energy drops from 0.96 ± 0.18 J to $\mathbf{0.71 \pm 0.51}$ J (**-26.0%**). Average package power decreases from 28.71 ± 2.49 W to $\mathbf{22.75 \pm 1.32}$ W (**-20.8%**), driving Energy-Delay Product down to $\mathbf{0.42 \pm 0.02}$ J·s (**-27.6% EDP reduction** vs. CFS's 0.58 ± 0.19 J·s). By confining communicating thread pairs to local cores and eliminating cross-CCX IPI wakeups, OPTIMA minimizes dynamic uncore and interconnect power dissipation.
- **Process Churn & Race-to-Sleep Optimization:** In `hackbench`, OPTIMA reduces package energy from 6.73 ± 0.78 J down to $\mathbf{4.12 \pm 0.18}$ J (**-38.8%**), slashing EDP by **65.3%** (1.01 ± 0.23 J·s on CFS vs. $\mathbf{0.35 \pm 0.04}$ J·s on OPTIMA). While instantaneous power is modestly higher (48.05 W vs. 45.34 W), OPTIMA's zero-lag local enqueue enables processes to finish in nearly half the time (0.086 s vs. 0.148 s in this isolated profile), achieving substantial race-to-sleep energy savings. (Note: this profile evaluated an isolated process group to isolate switching energy, while Table 2 reports the 16-group suite at 0.29 ± 0.01 s vs. 0.35 ± 0.04 s).
- **Batch Compute and Fixed-Duration Nuances:** In `sysbench cpu` (executed as a 5.0 s fixed-duration run), package power across all schedulers converges within $\approx$ 1–2% of CFS (35.90 W vs. 36.14 W), yielding identical package energy (179.68 J vs. 180.89 J). Because this benchmark operates over a fixed temporal window rather than fixed work, the reported EDP (899.28 J·s vs. 905.39 J·s) scales directly with energy. Evaluating energy per completed CPU event reveals the true trade-off: CFS retires 11,712 EPS consuming 3.09 mJ/event, whereas OPTIMA retires 6,028 EPS consuming 5.96 mJ/event due to density-based E-core steering and BPF dispatch pacing. Conversely, in `sysbench memory`, OPTIMA delivers 12.8% higher data throughput at energy parity (138.60 J vs. 137.69 J, +0.7%), confirming that memory saturation is preserved without energy penalty.

6.7 Limitations and Future Work

Several open challenges remain for extensible asymmetric scheduling: (1) *Cross-Platform Validation:* Expanding empirical evaluation beyond AMD Zen 5/5c silicon to Intel hybrid processors with HFI and ARM big.LITTLE / DynamIQ SoCs; (2) *Dynamic Power Integration:* Integrating density placement with Energy-Aware Scheduling (EAS) models and DVFS governors to throttle core frequencies during low-density phases; and (3) *Dynamic Burst Estimation:* Incorporating compiler-assisted runtime hinting or BPF quantile tracking to refine burst predictions in JIT-compiled runtimes.

7 Conclusion

In this paper, we presented **OPTIMA**, an extensible operating system scheduler for asymmetric multiprocessors in Linux `sched_ext` eBPF. By replacing empirical fair-share heuristics with formal combinatorial optimization paradigms—Smith's Rule WSPT density ordering, Online Admissible Preemption Filtering (OAPF) with adaptive peer hysteresis, density-guided domain placement, and synchronous IPC zero-lag dispatch—OPTIMA provides principled scheduling foundations. Rigorous isolated empirical evaluation across 16 workloads demonstrates a 6.3× surge in IPC throughput, an 84.1% reduction in pipe latency, a 94.6% reduction in worst-case latency spikes, and zero audio drops. Our transparent analysis characterizes workload trade-offs and trampoline overheads under process churn, establishing a principled empirical foundation for future asymmetric multiprocessor schedulers.

References

[1] Luca Abeni and Giorgio Buttazzo. Integrating multimedia applications in hard real-time systems. In *Proceedings of the 19th IEEE Real-Time Systems Symposium (RTSS)*, pages 4–13. IEEE, 1998.

[2] Sergey Blagodurov, Sergey Zhuravlev, Mohammad Dashti, and Alexandra Fedorova. A case for NUMA-aware contention management on multicore systems. In *Proceedings of the 2011 USENIX Annual Technical Conference (USENIX ATC)*, pages 1–15, Portland, OR, USA, 2011.

[3] John Bruno, Edward G. Coffman, and Ravi Sethi. Sequencing problems with partial ordered tasks and alternative processors. *Journal of the ACM*, 21(4):667–676, 1974.

[4] Quan Chen, Minyi Guo, and Haibing Guan. LAWS: Locality-aware work-stealing for multi-socket multi-core architectures. In *Proceedings of the 28th ACM International Conference on Supercomputing (ICS)*, ICS '14, pages 3–12, New York, NY, USA, 2014. ACM.

[5] Stijn Eyerman, Kristof Du Bois, and Lieven Eeckhout. Speedup stacks: Identifying scaling bottlenecks in multithreaded applications. In *Proceedings of the IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)*, pages 145–155. IEEE, 2012.

[6] Stijn Eyerman and Lieven Eeckhout. System-level performance metrics for multiprogram workloads. *IEEE Micro*, 28(3):42–53, 2008.

[7] Michael R. Garey and David S. Johnson. *Computers and Intractability: A Guide to the Theory of NP-Completeness*. W. H. Freeman and Company, San Francisco, CA, USA, 1979.

[8] Leonidas Georgiadis, Michael J. Neely, and Leandros Tassiulas. Resource allocation and cross-layer control in wireless networks. *Foundations and Trends in Networking*, 1(1):1–144, 2006.

[9] Haodong Guo, Quan Chen, Minyi Guo, and Liting Xu. SAWS: Selective asymmetry-aware work-stealing for asymmetric multi-core architectures. In *Proceedings of the IEEE 18th International Conference on High Performance Computing and Communications (HPCC)*, pages 116–123. IEEE, 2016.

[10] Leslie A. Hall, Andreas S. Schulz, David B. Shmoys, and Joel Wein. Scheduling to minimize average completion time: Off-line and on-line approximation algorithms. *Mathematics of Operations Research*, 22(3):513–544, 1997.

[11] Tejun Heo, David Vernet, Andrea Righi, and Changwoo Min. sched_ext: Bpf extensible CPU scheduling in the Linux kernel. In *Proceedings of the Linux Plumbers Conference (LPC)*, 2024. Linux 6.12 Mainline Implementation.

[12] Daniel Hodges. Scheduling at scale: eBPF schedulers with sched_ext. In *SREcon24 Europe/Middle East/Africa (SREcon24 EMEA)*, Dublin, Ireland, October 2024. USENIX Association.

[13] Jack Tigar Humphries, Neel Natu, Ashwin Chaugule, Ofir Weisse, Barret Rhoden, Josh Don, Luigi Rizzo, Oleg Rombakh, Paul Turner, and Christos Kozyrakis. ghOSt: Fast & flexible user-space delegation of Linux scheduling. In *Proceedings of the 28th ACM Symposium on Operating Systems Principles (SOSP)*, SOSP '21, pages 588–604, New York, NY, USA, 2021. ACM.

[14] Intel Corporation. Intel 64 and IA-32 architectures software developer's manual, volume 3b: Hardware feedback interface (HFI), 2021.

[15] Tsuyoshi Kawaguchi and Seiki Kyan. Worst case bound of an LRF schedule for the mean weighted flow-time problem. *SIAM Journal on Computing*, 15(4):1119–1129, 1986.

[16] Con Kolivas. The brain Fuck scheduler: BFS introduction. Web publication and Linux Kernel Archives, 2009.

[17] Con Kolivas. Multiple queue skiplist scheduler (MuQSS) for Linux. Web publication and Linux Kernel Archives, 2016.

[18] David Koufaty, Dheeraj Reddy, and Scott Hahn. Bias scheduling in heterogeneous multi-core architectures. In *Proceedings of the 5th European Conference on Computer Systems (EuroSys)*, EuroSys '10, pages 125–138, Paris, France, 2010. ACM.

[19] Rakesh Kumar, Keith I. Farkas, Norman P. Jouppi, Parthasarathy Ranganathan, and Dean M. Tullsen. Single-ISA heterogeneous multi-core architectures: The potential for processor power reduction. In *Proceedings of the 36th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO)*, pages 81–92. IEEE Computer Society, 2003.

[20] Rakesh Kumar, Dean M. Tullsen, Parthasarathy Ranganathan, Norman P. Jouppi, and Keith I. Farkas. Single-ISA heterogeneous multi-core architectures for multithreaded workload performance. In *Proceedings of the 31st Annual International Symposium on Computer Architecture (ISCA)*, pages 64–75. IEEE Computer Society, 2004.

[21] Juri Lelli, Claudio Scordino, Luca Abeni, and Dario Faggioli. Deadline scheduling in the Linux kernel. *Software: Practice and Experience*, 46(6):821–839, 2016.

[22] Jan Karel Lenstra, Alexander H. G. Rinnooy Kan, and Peter Brucker. Complexity of machine scheduling problems. *Annals of Discrete Mathematics*, 1:343–362, 1977.

[23] Giuseppe Lipari and Sanjoy K. Baruah. Greedy reclamation of unused bandwidth in constant bandwidth servers. In *Proceedings of the 12th Euromicro Conference on Real-Time Systems (ECRTS)*, pages 193–200. IEEE, 2000.

[24] Nicole Megow and Andreas S. Schulz. On-line scheduling to minimize average completion time revisited. *Operations Research Letters*, 32(5):485–490, 2004.

[25] Changwoo Min. scx_lavd: Latency-aware virtual deadline scheduling for high-framerate interactive computing. Technical report, Linux sched-ext Ecosystem and Igalia Technical Reports, 2024.

[26] Ingo Molnar. [PATCH] modular scheduler core and completely fair scheduler (CFS). Linux Kernel Mailing List (LKML), April 2007.

[27] Michael J. Neely. Energy optimal control for time-varying wireless networks. *IEEE Transactions on Information Theory*, 52(7):2915–2934, 2006.

[28] Michael J. Neely. *Stochastic Network Optimization with Application to Communication and Queueing Systems*, volume 3 of *Synthesis Lectures on Communication Networks*. Morgan & Claypool Publishers, 2010.

[29] Ricardo Neri. [PATCH v8 0/9] thermal/drivers/intel: Add support for intel hardware feedback interface. Linux Kernel Mailing List (LKML), 2022.

[30] Abhay K. Parekh and Robert G. Gallager. A generalized processor sharing approach to flow control in integrated services networks: The single-node case. *IEEE/ACM Transactions on Networking*, 1(3):344–357, 1993.

[31] Cynthia A. Phillips, Cliff Stein, and Joel Wein. Minimizing average completion time in the presence of release dates. *Mathematical Programming*, 82(1):199–223, 1998.

[32] Morten Rasmussen, Patrick Bellasi, Dietmar Eggemann, and Quentin Perret. Energy aware scheduling (EAS) in the Linux kernel. Linux Kernel Documentation, Documentation/scheduler/sched-energy.rst, 2018.

[33] Andrea Righi, Tejun Heo, David Vernet, and sched-ext contributors. sched-ext: scx_rlfifo and scx_rustland_core reference implementation and documentation. Linux Kernel sched-ext Repository, https://github.com/sched-ext/scx, 2024. Official sched-ext Round-Robin / FIFO user-space scheduler baseline based on scx_rustland_core.

[34] Juan Carlos Sáez, Manuel Prieto, Alexandra Fedorova, and Sergey Blagodurov. A comprehensive scheduler for asymmetric multicore systems. In *Proceedings of the 5th European Conference on Computer Systems (EuroSys)*, EuroSys '10, pages 139–152, Paris, France, 2010. ACM.

[35] Wayne E. Smith. Various optimizers for single-stage production. *Naval Research Logistics Quarterly*, 3(1–2):59–66, 1956.

[36] Ion Stoica and Hussein Abdel-Wahab. Earliest eligible virtual deadline first: A flexible and accurate mechanism for proportional share resource allocation. Technical Report TR-95-22, Department of Computer Science, Old Dominion University, Norfolk, VA, USA, 1995.

[37] Ion Stoica, Hussein Abdel-Wahab, Kevin Jeffay, Sanjoy K. Baruah, Scott E. Gehrke, and C. Greg Plaxton. A proportional share resource allocation algorithm for real-time, time-shared systems. In *Proceedings of the 17th IEEE Real-Time Systems Symposium (RTSS)*, pages 288–299. IEEE, 1996.

[38] Sergey Zhuravlev, Sergey Blagodurov, and Alexandra Fedorova. Addressing shared resource contention in multicore processors via scheduling. In *Proceedings of the 15th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, pages 129–142. ACM, 2010.

[39] Peter Zijlstra. sched: Earliest eligible virtual deadline first (EEVDF). Linux Kernel Mailing List (LKML), 2023. Merged in Linux Kernel 6.6.

A Artifact Appendix

A.1 Abstract

This artifact contains the complete source code, kernel eBPF bytecode, Rust topology controller, and automated evaluation harnesses for **OPTIMA**. It enables reproduction of all experimental results in this paper, including the 16-workload isolated benchmark suite (Table 2), hardware RAPL energy measurements (Table 4), and the synchronous IPC ablation study (Table 3).

A.2 Artifact Check-List (Meta-Information)

- **Program:** OPTIMA (Rust user controller + clang-compiled kernel BPF).
- **Compilation:** Rust 1.75+ (Cargo), Clang/LLVM 16+, bpftool.

- **Binary:** Release ELF executable (`scx_optima`).
- **Hardware:** Asymmetric multiprocessor with Performance and Efficiency cores. Primary evaluation: AMD Ryzen AI 9 HX 370 (Zen 5/5c: 4 P-cores @ 5.16 GHz + 8 E-cores @ 3.29 GHz, 12 cores, 24 threads). Compatible with Intel Hybrid (12th–14th Gen) and ARM big.LITTLE / DynamIQ (RK3588).
- **Run-time State:** Linux 6.12+ / 7.0+ with `CONFIG_SCHED_CLASS_EXT=y` and hardware RAPL energy monitoring.
- **Execution Time:** $\approx$ 1 minute for quick smoke test; $\approx$ 3.5 minutes for 5-repetition RAPL energy profiling; $\approx$ 20 minutes for full multi-run benchmark suite.
- **Output:** Console logs, structured JSON logs, Markdown reports, and LaTeX tables with Mean $\pm\sigma$ error bars.

A.3 Description & How to Access

The artifact is accessible via the anonymized research repository:
https://anonymous.4open.science/r/scx_optima-DAA.

A.4 Installation & Environment Setup

To verify the operating system prerequisites and compile the scheduler:

```
$ ./artifact/setup_env.sh
```

This script checks kernel `sched_ext` readiness, confirms hardware RAPL sysfs paths, verifies toolchain binaries, and compiles `target/release/scx_optima`.

A.5 Experiment Workflow

1. **Smoke Test (¡ 1 min):**

   ```
   $ ./artifact/run_artifact_eval.sh smoke
   ```

   Validates scheduler attachment to `sched_ext` and measures baseline vs. OPTIMA IPC pipe latency (16.06 μs $\rightarrow$ 2.55 μs).

2. **Hardware RAPL Energy Profiling:**

   ```
   $ ./artifact/run_artifact_eval.sh rapl
   ```

   Runs 5 independent executions per workload across CFS, `scx_optima`, `scx_rusty`, and `scx_rlfifo`, computing package/core energy, power, and Energy-Delay Product (Table 4).

3. **Viewing 34-Metric Benchmark Summary with Error Bars:**

   ```
   $ python3 artifact/parse_results.py
   ```

   Summarizes Table 2 with Mean $\pm\sigma$ across all 34 evaluated metrics.